XLC- CAPTUR

TEAM DESCRIPTION PAPER

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Team Introduction

Founded in 2009, XLC has been dedicated to advancing the field of robotics through innovation and teamwork. Since 2021, XLC Captur has continued this mission with our talented team members, each bringing unique skills and expertise to our collective effort.

Ondrej Peter: Our electrical designer, Ondrej, is the backbone of our team's electronic systems. His meticulous designs ensure our robots operate with precision and reliability.

Martin Šuriak: Martin does 3D modeling and mechanical design, and helps with everything. He creates our mechanical parts with designs that are functional and innovative, pushing what our robots can do.

Jakub Bohunický: Jakub, our programming expert, brings our robots to life with his great coding skills. His software is the brain of our robots, making sure they perform well in competitions.



Figure 1: Our team photo

Design overview

We began designing the first generation of our robot in late 2022 to compete in our home country's 2023 RoboCup Junior competition. Since then, we have developed many new skills, tackled design challenges, and learned from our failures.

Last year's competition turned out great for us, as we achieved 3rd place despite our robot's flawed designs, numerous malfunctions, and our initial struggles with coding.

This year, we have worked extremely hard and developed a new design for our top PCBs. We switched microcontrollers, added new sensors for IR ball detection and a camera, upgraded the gyro sensor, and integrated a kicker and dribbler.

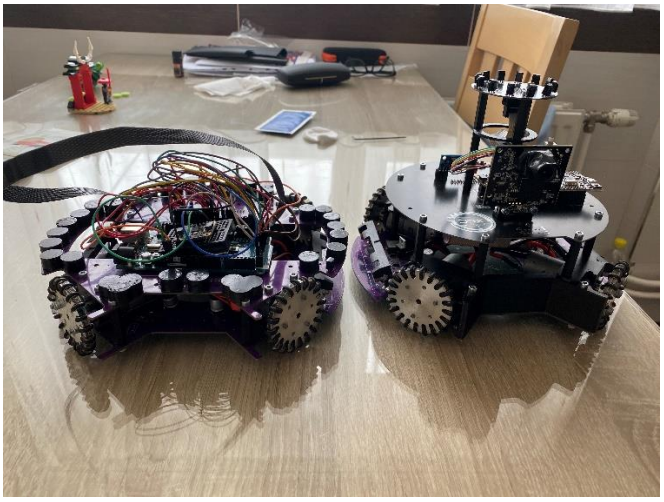


Figure 2 redisign of our robot

Mechanical design:

First, we began designing our robot as a 3D model in Fusion 360. This proved to be an excellent tool because it allowed us to create the robot according to the specified dimensions outlined in the competition rules. We decided that the main structure of our robot would be made of PCBs, and 3D modeling enabled us to seamlessly convert the robot's shape in Fusion 360 directly into PCB designs in EAGLE.

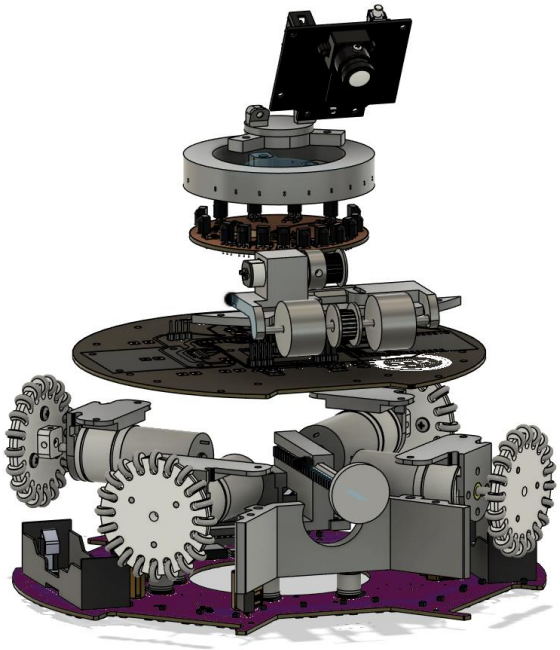


Figure 3: 3D model for current version of our robot

Kicker:

Our testing phase started with solenoids we had on hand, along with a DC motor that we adapted to move linearly using a small amount of 3D printing.

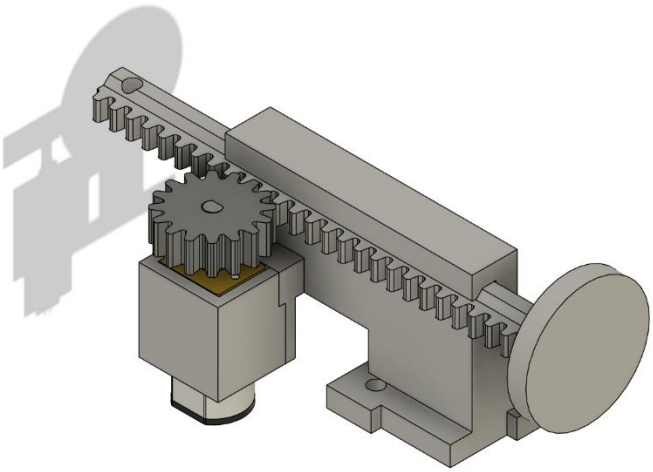


Figure 4: 3D model of our kicker

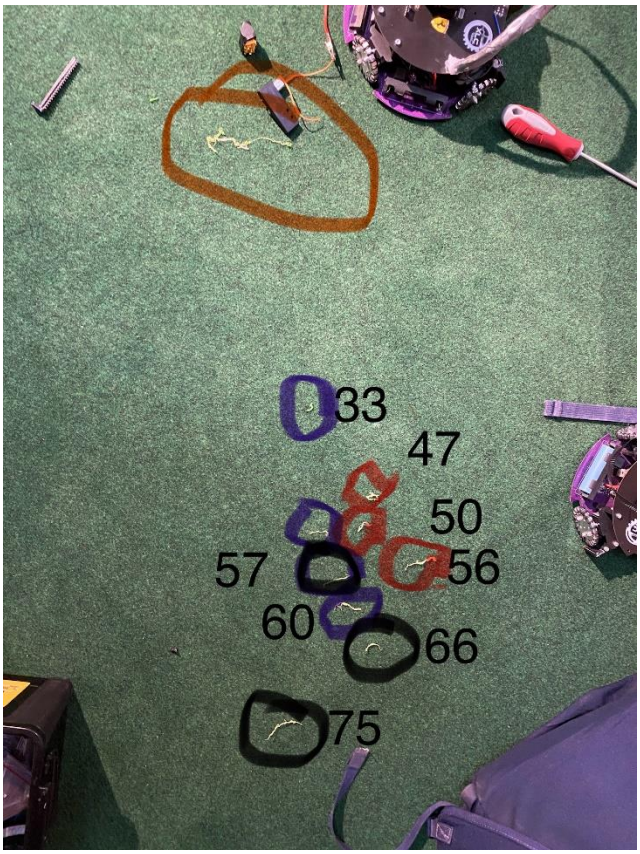


Figure 5: Our testing: red and purple are solenoids and black is our linear DC motor

Dribbler:

The dribbler consists of a DC motor connected by a GT2 belt to a rubber roller from a printer paper feeder, chosen for its softness and good adhesion properties. This rubber roller serves as the contact point for the ball, causing it to spin.

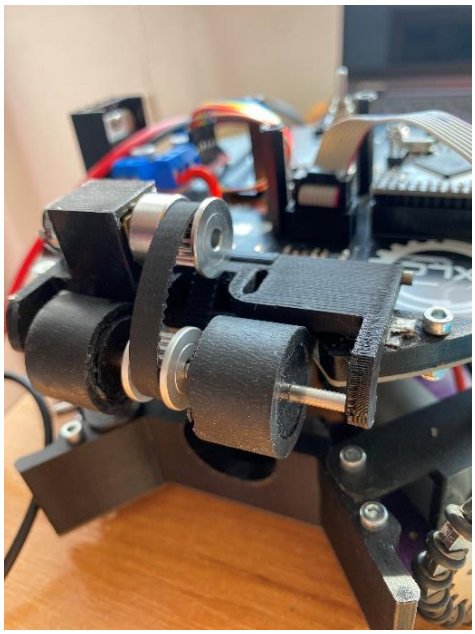


Figure 6: Dribbler with rubber rollers

To detect whether the ball is in the dribbler, we use a mechanical contact consisting of two parts of desoldering wire. When the dribbler makes contact with the ball, it lifts up and separates the contacts.

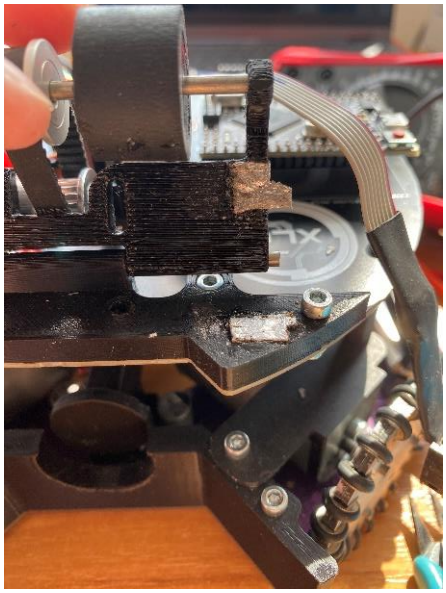


Figure 7: Ball detection switch

Homemade omniwheels:

We enlisted a trusted manufacturer to produce the wheel outlines and holes according to our specifications. After receiving the wheels, we carefully refined their surfaces through light sanding. Following this, we sourced axles and identified small wheels suitable for our needs. Lastly, we mounted these small wheels onto the axles and positioned them between the previously cut wheel outlines, securing everything in place with screws.



Figure 8 Building omniwheels

Electrical design:

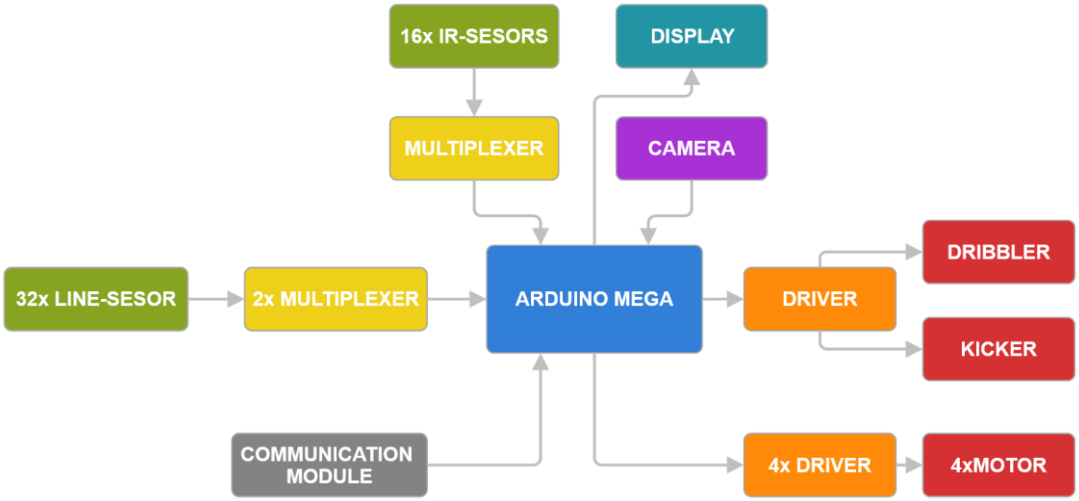


Figure 9: Logic block diagram

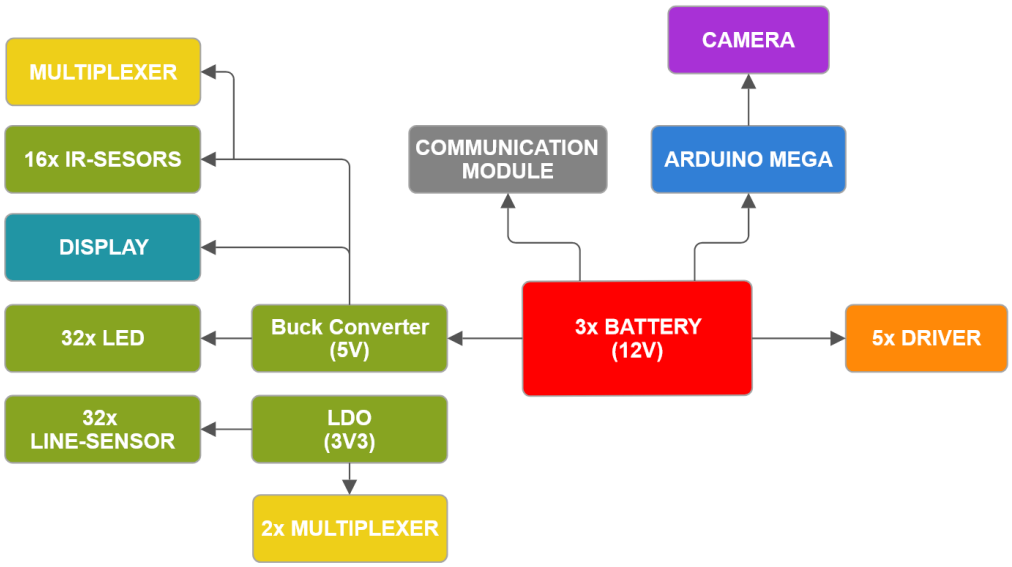


Figure 10: Power block diagram

Line Sensors: To detect the field borders, we use phototransistors that sense reflected light from the field that is emitted by LEDs. When our robot moves over a white border line, the sensors detect a change in voltage.

Use of Multiplexers: Given the limited number of analog inputs on the Arduino Mega, we use multiplexers for the line and IR sensors to expand our input capacity.

Camera: We utilize the Pixy Cam V2, which is connected to the Arduino via ISCP pins.

Motor Drivers: we pick VNH2SP30 motor drivers for simplicity of using PWM and their power output.

Battery: We choose a three lithium-ions cells in 18650 package because of compactness.

IR sensors: To track ball with most accuracy, we use sixteen 38khz IR sensors with lowpass filters.

Problems with electrical design:

The primary challenge we encountered involved ball detection. Initially, we lacked reference materials on reliable methods to detect the ball's location. We experimented with various solutions, and one breakthrough came when we switched our TSOP sensors from a 40kHz detection frequency to 38kHz TSSP sensor. This decision seemed counterintuitive because the ball emits IR pulses at 40kHz. However, adjusting the frequency helped mitigate issues with the sensors picking up reflections from different surfaces, which previously hindered accurate ball angle processing. Subsequently, through discussions with other teams, we integrated a low-pass filter into the sensor output, effectively reducing minor errors.

In our updated PCB designs, we opted to transition from Teensy 4.1 to Arduino Mega. This change was driven by our need for reliable and straightforward communication, particularly with motor drivers. The Teensy 4 microcontrollers operate at 3.3V, whereas our motor drivers

use 5V logic, necessitating logic voltage converters that did not perform as expected. Working with the Arduino Mega has proven simpler and more dependable in our setup.



Figure 9 arrival of our PCBS

Software design:

The robot is controlled by an Arduino Mega unit, which provides sufficient power and an adequate pin count. It is programmed in the Arduino IDE environment. The entire program is divided into various functions, which are executed in a specific order based on inputs and outputs.

First, the robot detects if it is off the line. Using a circular array of sensors and goniometric functions, it accurately determines its position and the direction it needs to move to stay on the line. If the robot is on the line, it then checks for the presence of a ball. If the ball is within the capturing zone, the robot uses a camera to locate the goal and move towards it. If the ball is not detected in the capturing zone, the robot searches for it using IR sensors mounted on top. If the IR sensors also cannot detect the ball, the robot remains stationary.

Throughout the program, the robot continuously monitors its deviation from the goal to minimize it. It also keeps track of the position of switches and the status of the communication module, pausing or recalibrating the gyro sensor as necessary.

Problems with programming:

During our programming efforts, we identified issues with the indexing of both TSSP sensors and phototransistors. These problems affected the accurate identification and operation of the components. After a thorough investigation, we pinpointed the root cause and implemented a solution, ensuring proper indexing for both types of sensors.

We also faced a problem with the Pixy v1.6 camera, which was transmitting a very low number of frames. This significantly impacted our system's performance and reliability. To resolve this,

we switched to the Pixy v2.1 camera, which provided a higher frame rate and improved overall functionality. The upgrade fixed the issue and ensured better performance for our application.

Furthermore, we discovered irregular input from the light sensors. The voltage output was inconsistent under identical lighting conditions, varying from one sensor to another. This affected the accuracy and reliability of our readings. After extensive testing and analysis, we identified the root cause and calibrated the sensors, ensuring uniform output across all light sensors in the same lighting conditions.



Figure 10: Testing our robot

BOM

	PCS	PRICE[€]
Arduino Mega	1	20
Motor drivers VNH2SP30	4	30
Motor Pololu 25D	4	20
Motor Kicker/Dribbler 1000rmp	2	5
Kicker and dribbler driver L298N	1	5
homemade OMNIWHEELS	4	0
PCBs	3	80
Gyro BNO055	1	50
Pixy V2.1	1	80
IR-Sensor TSSP58038	16	20
Other electronic components		40
Total		350

Figure 11: BOM per robot